

Introduction

This doc specifies how Programs and Exercises are modeled in Virtus.

Relationships

Programs have a number of Weeks. Each week has a number of Sessions. Each Session has a number of Exercises.

An Exercise has a number of Sets. A Set is defined by a number of Reps at a particular Intensity. Reps can be specified as a single number ("Reps"), a range ("Reps Range"), as many reps as possible ("AMRAP") or a time duration in minutes ("Time"). Intensity can be defined by rate of perceived exertion 0–10 ("RPE"), a percentage of the user's one repetition maximum ("% of 1RM"), a fixed increase from last week ("Last Week + X lbs") or a fixed increase from last session ("Last Session + X lbs").

Behaviors

Creating Template Programs

When the user or the Coach templatizes an Exercise as part of a custom Program, they define the number of Sets, the type of Reps to use for those Sets ("Reps", "AMRAP", etc.), and the type of Intensity to use for those Sets ("RPE", etc.). By default, all the Sets within an Exercise use the same Rep and Intensity types, but any individual Set can be overridden in an advanced menu (e.g. to make the last Set in an Exercise "AMRAP"). These act as targets for the user to reach within a session, e.g. "8 reps at 70% of 1RM".

Instantiating Sessions from Templates

Users can instantiate any Session within a Program to start a workout. Each Exercise in the Session can be filled out by the user to record their workout. The columns for an Exercise are as follows: Set (static), Intensity (static), Target (static), lbs (user input), Reps (user input), RPE (user input). Set is the set number. Intensity tells the user about the difficulty of the lift (e.g. "8 RPE", "85%", "LW +5", "LS +5"). Target is a summary of what the user should attempt to lift (e.g. "225 x 5"). Lbs is a number field for the user to input the actual weight lifted. Reps is a field for the user to input the actual reps completed. RPE is a number field for the user to input the actual RPE felt (1–10).

The data in the Target column needs to be computed based on each user. At the Profile level we will store the user's best lifts for each exercise from which we can compute any nRM (the most weight the user can lift for n reps) or RPE m (the weight and reps required to achieve m level of exertion). You'll need to come up with a formula to convert historical data to nRM and RPE m. Remind me to provide more robust formulas later. If there isn't any historical data, use a 1RM and RPE 10 of 100 lbs for every lift. In the future there will be an

onboarding experience for the user to set the 1RM and RPE 10 for every lift needed in a Session.

Creating Blank Sessions

Users may create blank Sessions to record one-off workouts from any collection of Exercises they want. The only columns are Set, lbs, Reps, and RPE. In this context it doesn't make sense to provide Intensity or Target since the Session isn't part of a Program. However, there will be an option within a blank Session to ask the Coach to provide guidance which may populate Intensity and Target columns. The level of guidance the Coach provides is up to the user. The user can specify the Exercises and the number of Sets, and the Coach can add and populate Intensity and Target columns. The Coach can also provide the number of Sets, or even suggest the Exercises themselves too (effectively generating a whole one-off Session). The Intensity and Target should be modulated based on user feedback (e.g. "Give me a light/typical/hard workout today") and historical data to provide the user with a workout that feels appropriate for them.

Instantiating Sessions from Previous Sessions

Users may want to repeat a previous workout they did, especially from one-off workouts. The details of how this works aren't clear to me right now. We'll solve this later. For now, understand that we need the ability to duplicate previous Sessions so the user can repeat a workout.